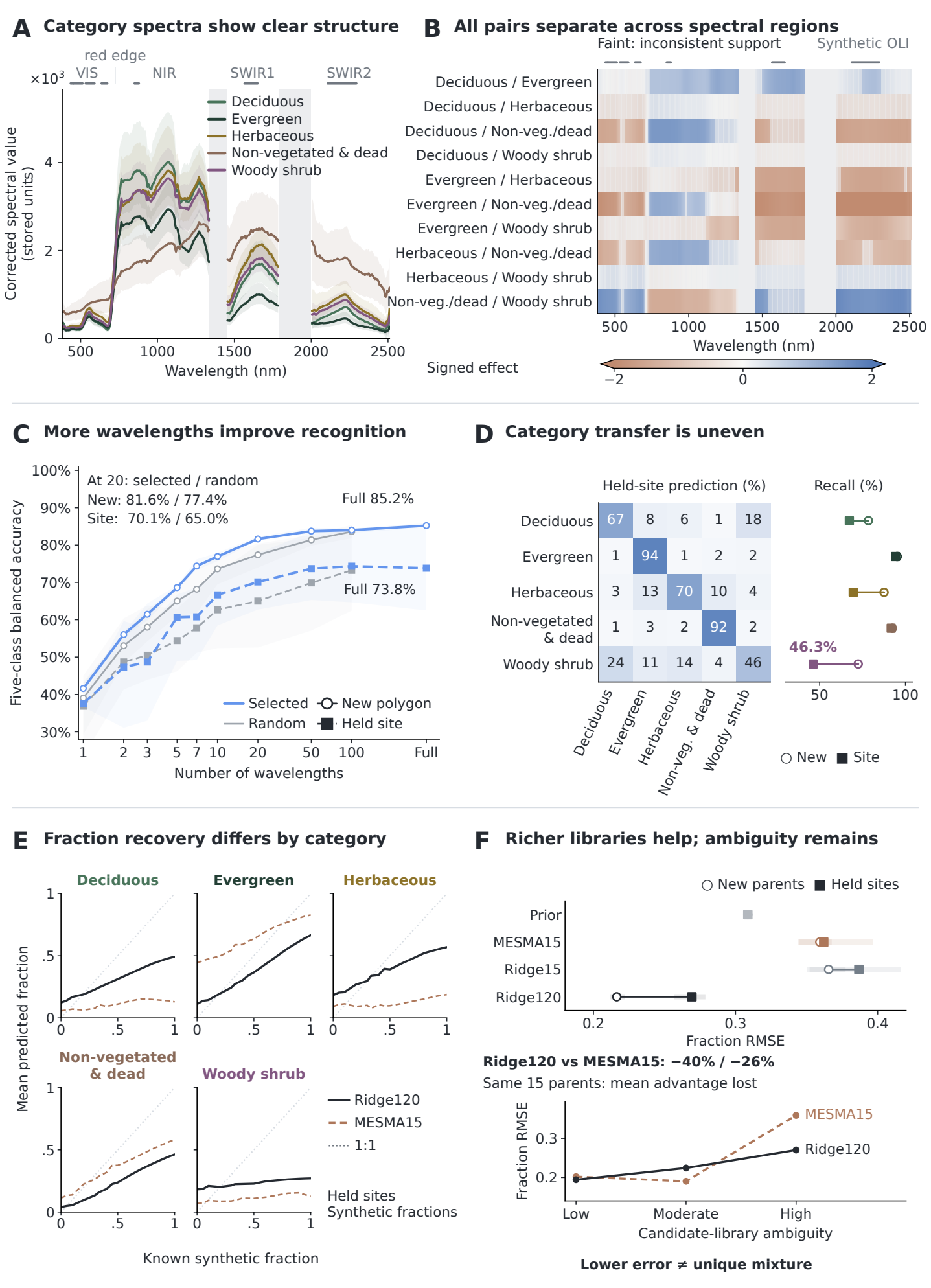

A
Category spectra show clear structure
red edge
VIS
NIR
SWIR1
SWIR2
×10³
Deciduous
Evergreen
Herbaceous
Non-vegetated & dead
Woody shrub
Corrected spectral value (stored units)
Wavelength (nm)
B
All pairs separate across spectral regions
Faint: inconsistent support
Synthetic OLI
Deciduous / Evergreen
Deciduous / Herbaceous
Deciduous / Non-veg./dead
Deciduous / Woody shrub
Evergreen / Herbaceous
Evergreen / Non-veg./dead
Evergreen / Woody shrub
Herbaceous / Non-veg./dead
Herbaceous / Woody shrub
Non-veg./dead / Woody shrub
Wavelength (nm)
Signed effect
C
More wavelengths improve recognition
At 20: selected / random
New: 81.6% / 77.4%
Site: 70.1% / 65.0%
Full 85.2%
Full 73.8%
Five-class balanced accuracy
Selected
Random
New polygon
Held site
Number of wavelengths
D
Category transfer is uneven
Held-site prediction (%)
Recall (%)
Deciduous 67 8 6 1 18
Evergreen 1 94 1 2 2
Herbaceous 3 13 70 10 4
Non-vegetated & dead 1 3 2 92 2
Woody shrub 24 11 14 4 46
46.3%
New
Site
E
Fraction recovery differs by category
Deciduous
Evergreen
Herbaceous
Non-vegetated & dead
Woody shrub
Mean predicted fraction
Ridge120
MESMA15
1:1
Held sites
Synthetic fractions
Known synthetic fraction
F
Richer libraries help; ambiguity remains
New parents
Held sites
Prior
MESMA15
Ridge15
Ridge120
Fraction RMSE
Ridge120 vs MESMA15: −40% / −26%
Same 15 parents: mean advantage lost
Fraction RMSE
MESMA15
Ridge120
Low
Moderate
High
Candidate-library ambiguity
Lower error ≠ unique mixture